

Assignment 2 – Case Study

Stage 4 Tutorial Activities

Object-Oriented Design Decisions

Week 7 | 60 minutes

Activity 1 - Encapsulation Review

Class	Protected state / invariant	Public operations
Patient	Patient name and ID, contact details	Register patient, update details, view information about the patient
Practitioner	Practitioner name and ID, specialty	Add practitioner, update details, view information about the practitioner
Appointment	Date, time, Appointment number, patient ID and Practitioner ID	Booking an appointment, cancel or update or reschedule an appointment

Activity 2 - Composition or Inheritance?

Appointment and Patient -> ☐ Composition/association Reason: Appointment is associated with patient and patient is not a category or type of appointment.

Appointment and Practitioner -> ☐ Composition/association Reason: Appointment is associated with practitioner and patient is not a category or type of appointment.

Doctor and Practitioner (hypothetical) -> Inheritance Reason: A doctor can be a practitioner

Clinic and Appointment -> ☐ Composition/association Reason: A clinic organizes the appointments.

Activity 3 - Responsibility Allocation

Who decides whether SCHEDULED can become CANCELLED?

- Appointment class because it has the authority to set rules and owns the appointment status.

Who validates a patient name?

- Patient class should ensure that the given name is valid.

Should Appointment execute SQL? Why?

- No. Appointment should handle appointment and rules not database operations.

Should the UI decide whether a status transition is legal?

- No. Only the appointment class should be doing this so the business rules are protected.

Activity 4 - AI Code Critique

AI generates an Appointment class with public status mutation, SQL inside cancel(), a NotificationManager dependency and inheritance from PatientRecord. Identify at least five design problems and corrections.

Problems	Corrections
1) Appointment may have many responsibilities	Keeping appointment strict to business rules and appointment data
2) Appointment might be dependent on manager	Using a completely separate service mechanism
3) Patient record file may effect the appointment	Remove data inheritance from patient record file
4) Status of appointment may be invalid	Appointment should enforce validation of status
5) Public showing of status allows anything to change the appointment	Confine the status to private and allow controlled editing

Exit question

Why can code be object-oriented syntactically but still have poor object-oriented design?

- Object oriented programming is a way of designing software by organizing it around objects rather than just functions but using classes and objects does not necessarily mean the design will be good. It can turn out inefficient if the classes have too many responsibilities, encapsulation is weak and inheritance is used incorrectly. A good object-oriented design means responsibilities, data, and relationships are organized correctly.